

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: _____ ZHOU Ke _____

Student ID: _____ 23099817D _____

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_____ show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129

Does R2 show up as an OSPF neighbor on R1? __ Yes _____

Does R2 show up as an OSPF neighbor on R3? __ No _____

What does this information tell you?

__ Traffic destined for the 192.168.2.0/24 network from R3 is routed through R1. Because the S0/0/1 interface on R2 remains configured as passive, OSPF does not advertise routing information across that link. The total accumulated cost of 129 arises from the path taken: packets must traverse two T1 serial connections (each with a cost of 64) and then cross R2's GigabitEthernet 0/0 LAN interface (with a cost of 1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

__ R2(config) # router ospf 1

__ R2 (config-router) # no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 (One T1 serial link (1.544 Mb/s at a cost of 64) plus the R2 Gigabit 0/0 LAN link (at a cost of 1))

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because modern networking equipment typically supports link speeds well above 100Mb/s. To ensure cost calculation accurately reflect these higher-speed connections, the default reference bandwidth must be increased.

Step 2: Change the bandwidth for an interface.

- g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 + R3 G0/0(781 + 1 = 782).

192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 (781+64 = 845)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562..Each serial link now carries a cost of 781. Since the path to the 192.168.23.0/24 network traverses two serial links, the total cost is calculated as 781 + 781 = 1562

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

__ OSPF selects the path with the lowest total cost. In this case, the preferred route is R1-S0/0/0 → R2-S0/0/1 → R3-G0/0, which has a combined cost of $781 + 781 + 1 = 1,563$. This is lower than the alternative path R1-S0/0/1 → R3-G0/0, which totals $1,565 + 1 = 1,566$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

__ Router ID assignments determine which devices become the Designated Router (DR) and Backup Designated Router (BDR) on a multi-access network. If the router ID is tied to an active physical interface, it may change if that interface goes down. To avoid this, the router ID should be set using a loopback interface address (which remains up) or explicitly configured with the **router-id** command.

2. Why is the DR/BDR election process not a concern in this lab?

__ The DR/BDR election process only applies to multi-access networks such as Ethernet or Frame Relay. Since the lab uses point-to-point serial links, no DR/BDR election takes place.

3. Why would you want to set an OSPF interface to passive?

__ Configuring a LAN interface as passive prevents unnecessary OSPF routing updates from being sent on that interface, conserving bandwidth. However, the router will still advertise the associated network to its neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

In summary, OSPF's hierarchical structure—achieved through the use of areas—enhances scalability by dividing large networks into manageable segments. This design limits the scope of OSPF operations, reduces computational overhead, and improves overall stability and performance. By organizing networks in this way, OSPF becomes highly effective for supporting very large and complex environments.

5. What single OSPF route ~~configuration command~~ *absent* allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

This command assigns all interfaces within the specified range to OSPF area 0, also known as the backbone area. Area 0 serves as the central hub that interconnects all other OSPF areas, making it essential for proper communication across the network. By applying this command, OSPF treats the defined IP address range as part of area 0, ensuring routing information is exchanged within that range and enabling seamless connectivity between devices across different areas.

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

In this scenario, you have three routes to the same network with the following metrics:

RIP route with a metric of 4.

OSPF route with a metric of 3,053,092

EIGRP route with a metric of 4,039,043.
